

Y7 Starters

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October 2023

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Abstract

A collection of assorted Year 7 worksheets. The self-assessment tasks in Summer HT1 are particularly good revision ahead of summer assessments.

1 Retrieval/Mixed Practice

1.1 Autumn HT1

1.2 Autumn HT2

1. Evaluate $-17+9$
2. Evaluate $-180 \div 12$
3. Simplify $\sqrt{20}$
4. Simplify $\sqrt{108}$
5. Evaluate $-3 + 9 \div 6 \times 2^2$
6. Evaluate $(7 - 12)^2 \times (3^3 - 17) \div 2$
7. Evaluate $\sqrt[3]{729}$
8. Evaluate $\sqrt{256}$
9. Find the prime factorisation of 84
10. Find the prime factorisation of 612

Starter

1. Sketch the line $y = x$
2. Sketch the line $x = -2$
3. Sketch the line $y = -x$
4. Sketch the line $y = 3$
5. Does the point $(-1, 1)$ lie on any of the above lines? If so, which line?
6. Does the point $(-3, 2)$ lie on any of the above lines? If so, which line?
7. Simplify $2(3 + x)$
8. Simplify $7(4 - y)$
9. Simplify $-3(z - 1)$
10. Simplify $x(12 - x)$
11. Simplify $-3y(1 + y)$
12. Simplify $-2z^2(3z - 5)$
13. Simplify $2x(x + 1) + 3(x - 4)$
14. Simplify $3x(1 - x) - 4x^2(1 + 2x)$

Starter

1. Solve $2x + 1 = 7$
2. Solve $9 - 2y = -4$
3. Solve $\frac{5z+2}{-4} = 5$
4. Evaluate $1\frac{5}{12} + \frac{2}{3}$
5. Solve $1 - 2(w - 4) = 5$
6. Evaluate $7\frac{1}{5} - 2\frac{5}{6}$
7. Evaluate $\frac{196}{25} \times \frac{65}{28}$
8. Evaluate $-2\frac{3}{5} \div 3\frac{3}{4}$

Answers

1. $x = 3$

2. $y = 6\frac{1}{2}$

3. $z = -3\frac{3}{5}$

4. $w = 2$

5. $2\frac{1}{12}$

6. $4\frac{11}{30}$

7. $18\frac{1}{5}$

8. $-\frac{52}{75}$

Autumn Term Review Homework

Solve the following questions. The solution to each has a corresponding letter given on the page overleaf. Solve each question to reveal the hidden message.

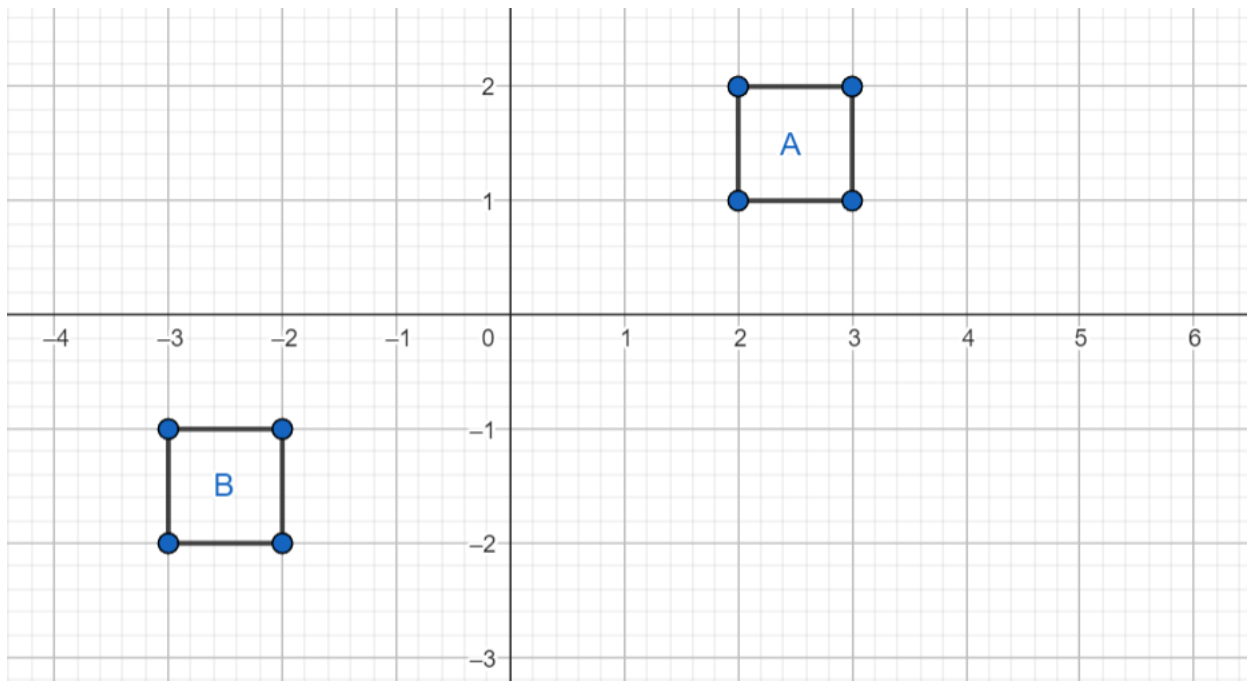
1. Evaluate $17 - 23 + 4$
2. Evaluate $(3 - 5)^3 \div 16 \times 7$
3. Evaluate $0.738 \div 0.18$
4. Evaluate $-3.94 + 8.04$
5. Simplify fully $\sqrt{72}$
6. Determine 0.0030721451 accurate to 3 significant figures.
7. Determine the index of 7 in the prime factorisation of 271656.
8. Evaluate 3.2×1.28125
9. Determine the coefficient of x^2 in the expansion of $7x(3 - x) - x^2(x - 1)$
10. Evaluate $-\frac{13}{9} + 1\frac{1}{3}$
11. Substitute $x = -2$ into the expression $3x^2 - 7x$
12. Determine the x -coordinate of the intersection between the lines $y = -x$ and $y = 2$?
13. An isosceles triangle has angles 54° and 62° . What is the missing angle?
14. Evaluate $\frac{3}{16} \times \left(\frac{1}{3} - 1\right)^3 \div \frac{1}{2}$

A	62
B	54
C	0.00307
D	$3\frac{1}{2}$
E	$-3\frac{1}{2}$
F	4.231
G	0.003
H	3
I	-6
J	$-1\frac{3}{7}$
K	$2\sqrt{18}$
L	8
M	-2
N	0.00
N	$3\sqrt{8}$
O	-8
P	0.003072
Q	-26
R	4.1
S	$-\frac{1}{9}$
T	26
U	π
V	$6\sqrt{3}$
W	6
X	2
Y	$6\sqrt{2}$
Z	$\frac{7}{12}$

1.3 Spring HT2

Retrieval Practice

1. Evaluate $2 + 12^2 \div 6$
2. Determine the prime factorisation of 1386.
3. Sketch a graph of $y = 4$ and $y = -x$. At which coordinate do they intersect?
4. Determine $\frac{7}{5} - 1\frac{2}{15}$
5. In the graph below, fully describe two transformations from A to B .



1.4 Summer HT1

Self-Assessment Task 1

1. Solve $3x - 7 = 59$.
2. Evaluate $15 - 6^2 \div 3$.
3. Plot the line $y = -3$ in Figure 1 below.
4. Evaluate $(-3)^3 \div (-2 + (-4))$ as an improper fraction.
5. Solve $5x - 20 = 4 - 3x$.
6. Plot the coordinates $(-1, 2)$, $(3, 4)$ and $(5, 0)$ on Figure 1 below.
7. What coordinate is now needed to form a rectangle in Figure 1?
8. Solve $\frac{4}{9}x - 2 = \frac{1}{3}(x + 1)$.
9. Evaluate $16 \div 8 \div (4 \div 2)$.
10. Evaluate $\frac{1}{1 + \frac{1}{1 + \frac{1}{3 + \frac{1}{4}}}}$.

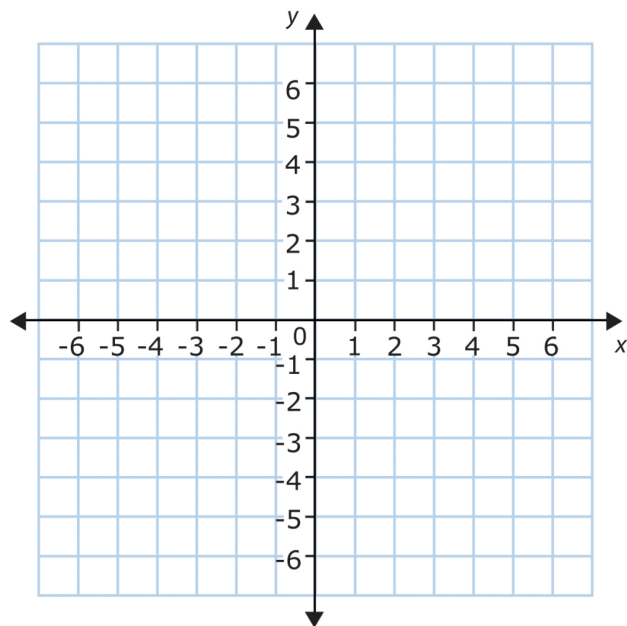
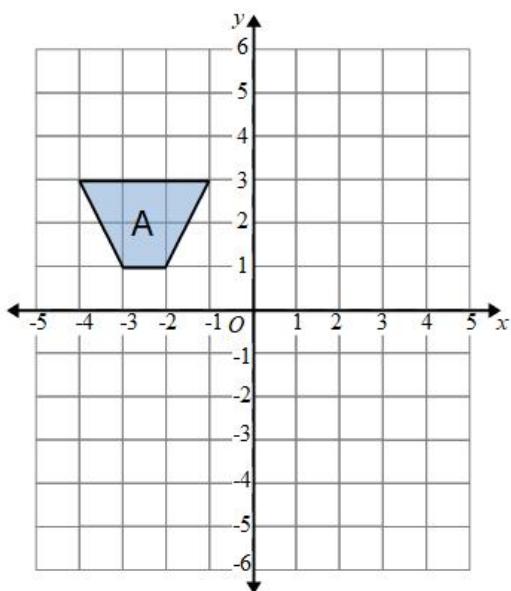


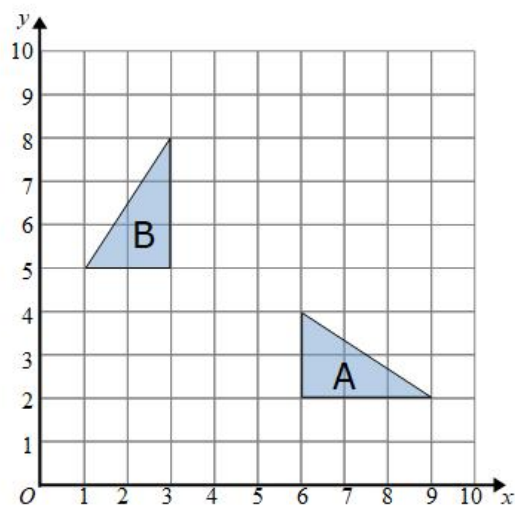
Figure 1

Self-Assessment Task 2

1. Evaluate 0.00321×0.014 .
2. Translate shape A in Fig. (a) by the vector $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$.
3. Evaluate $\frac{17}{19} + \frac{12}{57}$ as a mixed number.
4. Reflect shape A in Fig. (a) about the line $y = x$.
5. Evaluate $2.25 \div 0.0045$.
6. Describe fully the transformation from A to B shown in Fig. (b).
7. Evaluate $\frac{125}{27} \div 1\frac{2}{3}$ as a mixed number.
8. State 0.021004 to 3 significant figures.
9. Evaluate $-\frac{5}{9} + \left(-\frac{4}{5}\right)^2$.
10. A number after being rounded to 2 significant figures is 100. What is the lowest value this number could be?



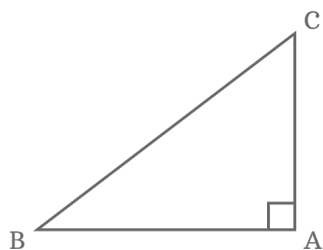
(a)



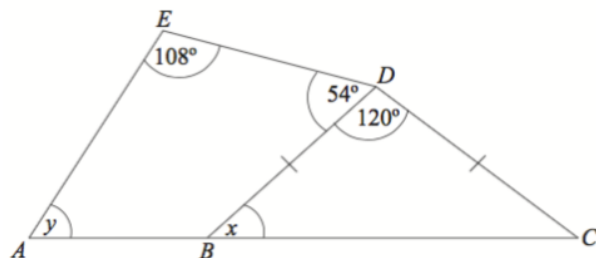
(b)

Self-Assessment Task 3

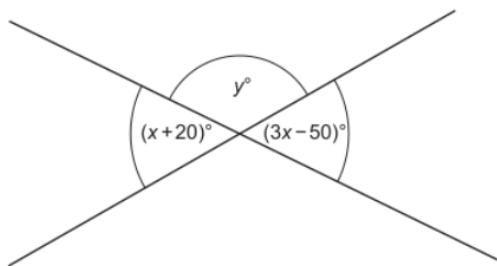
1. Simplify the following expression $3x(x - 4) + 5x$.
2. A triangle has side lengths of 2cm, 4cm and 6cm. Is this a right angled triangle? Explain your answer fully.
3. In the triangle shown below in figure (a), $AB = 2.4\text{cm}$ and $BC = 2.5\text{cm}$. Determine the length of AC .
4. Determine the size of angle x in figure (b) below.
5. Substitute $a = 2$ and $b = -0.5$ into the expression $\frac{a+b^2}{2ab}$.
6. Determine the size of angle y in figure (b) below.
7. Simplify the following expression $2xy^2 - 2xy + 2x^2y - 2xy + 2xy^2$.
8. Determine the size of angle y in figure (c). Give reasons for your answer.
9. Determine the distance between the points $(3, -2)$ and $(-2, -14)$.
10. Determine the length DE in figure (d) if the total area of $ABCED$ is 60 cm^2 .



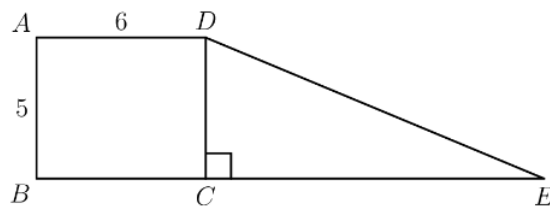
(a)



(b)



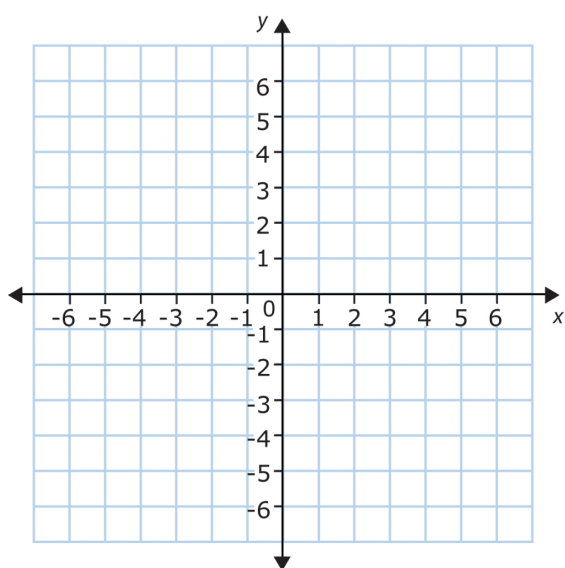
(c)



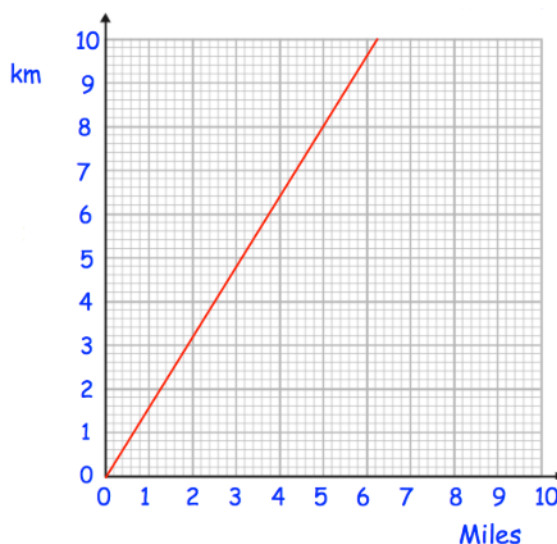
(d)

Self-Assessment Task 4

1. Determine the probability of obtaining two heads when a coin is flipped three times.
2. Plot the line $y = 2x - 2$ on graph (a) below.
3. Having plotted $y = 2x - 2$, state the equation of the line that would need to be plotted to graphically solve the equation $-3x + 4 = 2x - 2$
4. Convert 3.3125 into a fraction.
5. If a biased dice has a probability of 0.4 of landing on a 6, what is the expected number of sixes in 250 rolls?
6. Find 17.5% of 24.
7. A four sided spinner can land on any colour of: blue, red, green and yellow. The probability of landing on red is 0.2 and on blue is 0.35. The probability of landing on yellow is 14 times smaller than that of green. Determine the probability of landing on green.
8. Use graph (b) to convert 10 miles into km.
9. Evaluate $16.2\% - \frac{5}{8} \times 0.42$
10. In a bag there are 4 blue balls and x red balls. Adding 10 blue balls triples the probability of picking a blue. How many red balls are there?



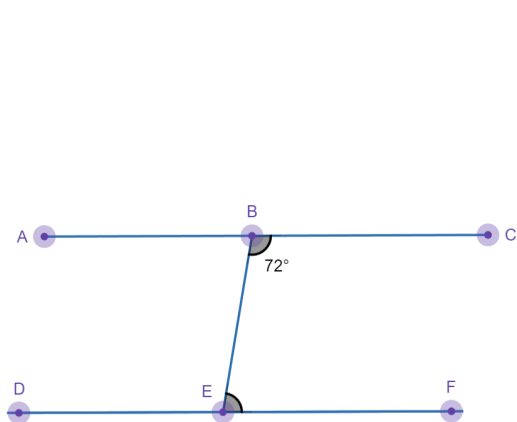
(a)



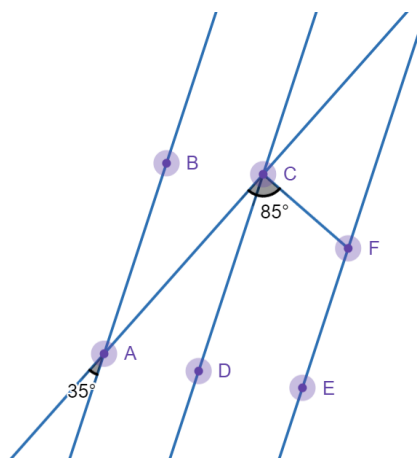
(b)

Self-Assessment Task 5

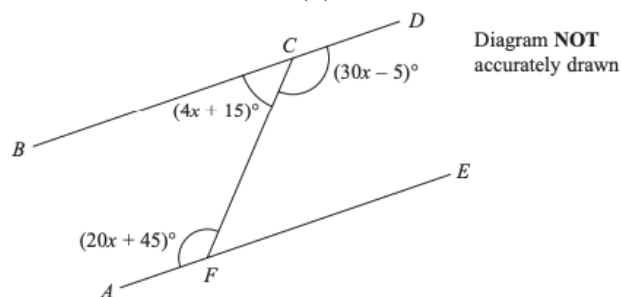
1. Determine the prime factorisation of 630.
2. Lines ABC and DEF are parallel in figure (a). Determine the size of angle BEF stating a reason for your answer.
3. Determine the prime factorisation of 13013.
4. Determine a formula for the n^{th} term of the sequence given by $3, -\frac{5}{2}, -8, -\frac{27}{2}, \dots$
5. Is 51 in the sequence defined by $u_n = 7 + 3n$. Justify your answer.
6. Determine, using prime factorisation, the square root of 5929.
7. A sequence is defined by $u_n = \frac{2n+1}{3-n}$. Determine the 5^{th} term in the sequence.
8. In figure (b) below, you are given that the lines AB , CD , and EF are parallel. Determine the size of angle CFE .
9. In figure (c), AFE and BCD are straight lines. Prove that they are parallel.
10. Determine whether or not 41075800 is a cube or not. Justify your answer.



(a)



(b)



(c)

Diagram **NOT**
accurately drawn

2 Fractions

Fraction Arithmetic

Give all answers as mixed numbers where appropriate.

1. $\frac{2}{7} + \frac{3}{5}$

2. $\frac{3}{4} - \frac{1}{6}$

3. $\frac{2}{5} \times \frac{9}{11}$

4. $\frac{13}{19} \times \frac{3}{26}$

5. $\frac{7}{8} \div \frac{1}{4}$

6. $3\frac{1}{4} + 2\frac{4}{5}$

7. $5\frac{5}{12} - 3\frac{2}{3}$

8. $2\frac{2}{15} \times 1\frac{1}{4}$

9. $3\frac{2}{3} \div 1\frac{2}{5}$

10. $\left(\frac{6}{7}\right)^3$

1. $\frac{31}{35}$
2. $\frac{7}{12}$
3. $\frac{18}{55}$
4. $\frac{3}{38}$
5. $3\frac{1}{2}$
6. $6\frac{1}{20}$
7. $1\frac{3}{4}$
8. $2\frac{2}{3}$
9. $2\frac{13}{21}$
10. $\frac{216}{343}$

3 Equations

Solving Equations with Fractions Starter

Solve the following equations:

1. $a + \frac{1}{2} = 3$

2. $b - \frac{3}{5} = -\frac{1}{4}$

3. $\frac{2}{3}c = 7$

4. $\frac{d}{7} = \frac{4}{11}$

5. $\frac{3w}{-4} = \frac{9}{16}$

6. $\frac{2}{5}x + 1 = \frac{1}{3}$

7. $\frac{3y}{7} - \frac{1}{4} = -2$

8. $\frac{-2}{9}z - \frac{2}{3} = \frac{5}{9}$

Answers

1. $a = 2\frac{1}{2}$

2. $b = \frac{7}{20}$

3. $c = 10\frac{1}{2}$

4. $d = 2\frac{6}{11}$

5. $w = -\frac{3}{4}$

6. $x = -1\frac{2}{3}$

7. $y = -4\frac{1}{12}$

8. $z = -5\frac{1}{2}$

Decimal Arithmetic and Solving Equations Starter

1. Evaluate the following:

(a) $0.231 + 0.0416$

(b) $0.0415 - 0.278$

(c) 0.245×3.1

(d) $7.29 \div 0.081$

2. Solve the following equations for x :

(a) $3x - 4 = -16$

(b) $3(x - 2) - 4(2x + 5) = 1$

(c) $-4x + 2 = 7x - 15$

(d) $\frac{2}{3}x - \frac{5}{6} = 5$

3. Two lengths of an equilateral triangle are $3x - 1$ and $x + 2$. Form and solve an equation in x .

1. (a) 0.2726
(b) -0.2365
(c) 0.7595
(d) 90
2. (a) $x = -4$
(b) $x = -5\frac{2}{5}$
(c) $1\frac{6}{11}$
(d) $x = 8\frac{3}{4}$
3. $x = 1\frac{1}{2}$

Fractions, Decimals and Percentages

1. Convert 0.825 into a fraction.
2. Convert 32.5% into a fraction in its simplest form.
3. Write 3.015 as a percentage.
4. Write $\frac{17}{25}$ as a decimal.
5. Find 93% of 320.
6. Find $\frac{3}{5}$ of 72. Give your answer as a decimal.
7. Increase 52 by 4%.
8. Decrease 36 by 12%.
9. If $\frac{3}{17}$ of a quantity is 33, find the original quantity.
10. Evaluate $0.23 - 16.5\% + \frac{5}{8}$ as a fraction.

1. $\frac{33}{40}$
2. $\frac{13}{40}$
3. 301.5%
4. 0.68
5. 297.6
6. 43.2
7. 54.08
8. 31.68
9. 187
10. $\frac{69}{100}$

4 BIDMAS, Powers, Surds and Negatives

Surds

Simplify the following surds:

1. $\sqrt{36}$

2. $\sqrt{169}$

3. $\sqrt{8}$

4. $\sqrt{20}$

5. $\sqrt{48}$

6. $\sqrt{\frac{16}{25}}$

7. $\sqrt{\frac{169}{144}}$

8. $\sqrt{1\frac{34}{81}}$

9. $\sqrt{6.25}$

10. $\sqrt{0.5625}$

1. 6
2. 13
3. $2\sqrt{2}$
4. $2\sqrt{5}$
5. $4\sqrt{3}$
6. $\frac{4}{5}$
7. $\frac{13}{12}$
8. $1\frac{5}{9}$
9. $\frac{5}{2}$
10. $\frac{3}{4}$

Surds

Simplify the following expressions:

1. $\sqrt{49}$

2. $\sqrt{196}$

3. $\sqrt{12}$

4. $\sqrt{45}$

5. $\sqrt{96}$

6. $\sqrt{\frac{9}{64}}$

7. $\sqrt{\frac{121}{324}}$

8. $\sqrt{\frac{7}{144}}$

9. $(\sqrt{10})^2$

10. $(3\sqrt{7})^2$

5 Pythagoras

Pythagoras Applications

1. Determine the distance between the following points. Give your answer as a simplified surd if necessary.
 - (a) $(3, 2)$ and $(9, 10)$
 - (b) $(3, -13)$ and $(-5, 2)$
 - (c) $(5, 2)$ and $(1, 4)$
 - (d) $(-1, -7)$ and $(5, -11)$
2. Determine two possible areas of an isosceles triangle with side lengths 3cm and 2cm.
3. A parallelogram has an area of 12cm^2 and one of the lengths is 8cm. Determine the remaining length.

1. (a) 10cm
(b) 17cm
(c) $2\sqrt{5}$ cm
(d) $2\sqrt{13}$ cm
2. $\frac{3\sqrt{13}}{8}$ or $\sqrt{2}$
- 3.

6 Straight Line Graphs

Plotting Straight Lines

Plot the following straight line graphs:

1. $y = x + 2 \quad \{0 \leq x \leq 4\}$

2. $y = x - 4 \quad \{-2 \leq x \leq 2\}$

3. $y = 2x \quad \{-4 \leq x \leq 4\}$

4. $y = -3x \quad \{-4 \leq x \leq 0\}$

5. $y = 2x + 1 \quad \{-2 \leq x \leq 2\}$

6. $y = 2x - 1 \quad \{0 \leq x \leq 4\}$

7. $y = 1 - 2x \quad \{-2 \leq x \leq 2\}$

8. $y = -5x - 2 \quad \{-4 \leq x \leq 0\}$

Coordinates and Equations of Lines

For this worksheet you should draw your own axes and add coordinates and lines as appropriate to the question.

1. A square has three vertices at $(1, 2)$, $(1, 7)$, $(6, 2)$.
 - (a) Determine the 4th vertex.
 - (b) State the equations of the lines passing through the edges of this square.
2. A rectangle has three vertices at $(-3, 5)$, $(-3, 12)$, $(2, 5)$.
 - (a) Determine the 4th vertex.
 - (b) State the equations of the lines passing through the edges of this rectangle.
3. An isosceles triangle has vertices at $(0, 0)$ and $(1, 1)$. One of its edges lies on the line $y = 0$. Determine the coordinates of its third vertex.
4. A trapezium has three vertices at $(-4, -1)$, $(2, 3)$, $(-2, 3)$. Its fourth vertex lies on the line $x = 5$. Determine the coordinates of the 4th vertex.
5. A trapezium is formed by the region bounded by the lines $x = -7$, $x = -3$, $y = x$ and the x -axis. Find the coordinates of the vertices.
6. A parallelogram is formed by the region bounded by the lines $y = -12$, $y = -7$, $y = -x$. One of its vertices has coordinate $(14, -12)$. Determine the other 3 vertices.

1. (a) $(6, 7)$
(b) $x = 1, x = 6, y = 2, y = 7$
2. (a) $(2, 12)$
(b) $x = -3, x = 2, y = 5, y = 12$
3. $(2, 0)$
4. $(5, -1)$
5. $(-7, 0), (-3, 0), (-7, -7), (-3, -3)$
6. $(12, -12), (7, -7), (9, -7), (14, -12)$